

SOLAR WEATHER

7 JUL 2026



Lewis Thompson
W5IFQ

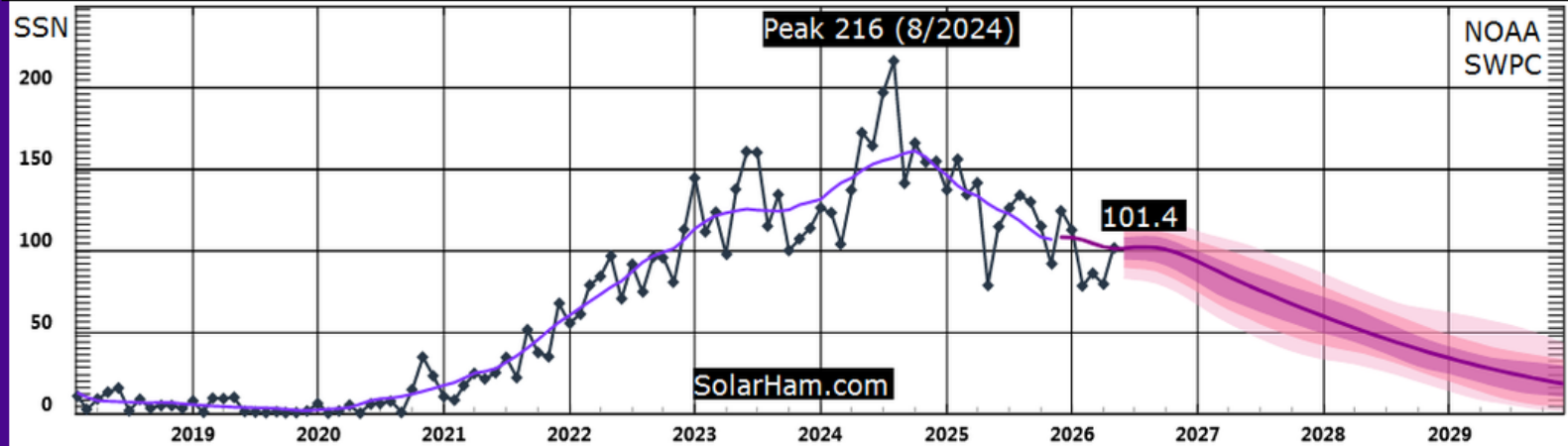
Taken by Jessica Fridrich on July 4, 2026
@ Binghamton, NY

Solar Cycle 25 Progression

(Updated June 7, 2026)

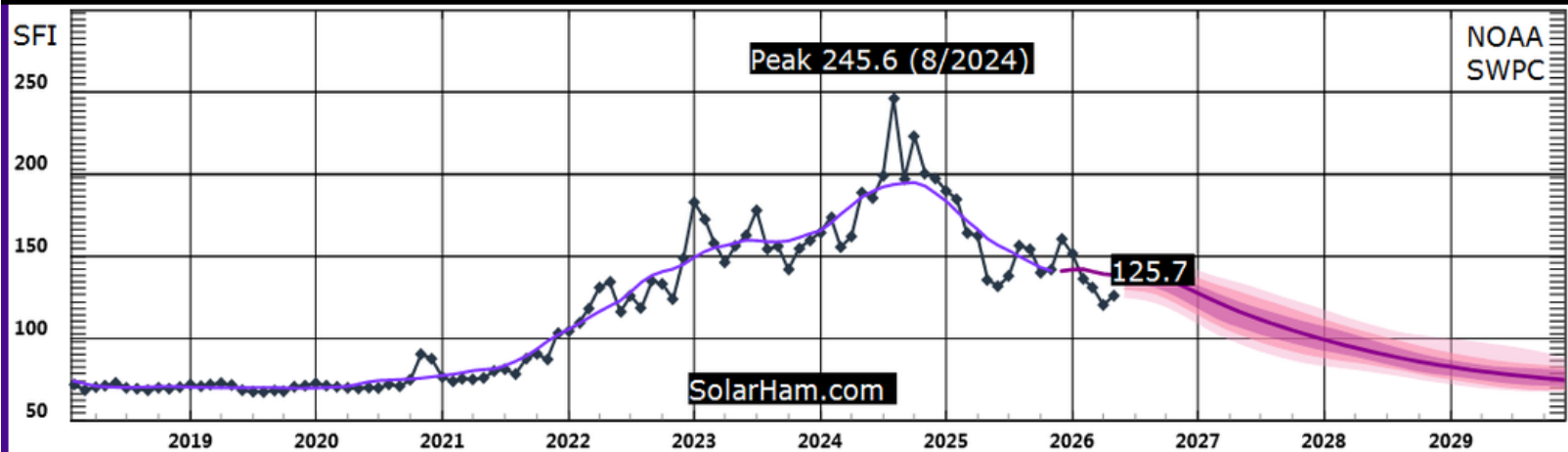
Sunspot Number Progression (May 2026)

Predicted SSN: **101.1** Actual: **101.4** Latest Smoothed Predicted SSN (11/2025): - Actual: **106.7**



10.7cm Solar Flux Progression (May 2026)

Predicted SFI: **138.3** Actual: **125.7** Latest Smoothed Predicted SFI (11/2025): - Actual: **140.8**



SolarHam

Indices: (7/7 @ 00:35 UTC)

SFI

125

▼ 14

SSN

85

3 Day Geomagnetic Forecast

July 7

July 8

July 9

2 (G0)

2 (G0)

3-4 (G0)

Max Kp

M-Lat 01% M-Lat 01% M-Lat 05%

H-Lat 20% H-Lat 15% H-Lat 20%

Probabilities

Latest SWPC Forecast (@ 00:30 + 12:30 UTC)

Visible Sunspot Regions

Sunspot Summary ►

SRS

AR 4485

B

S11W01

Stable

AR 4482

BG

S09E54

Growing

AR 4478

BG

S05W90

Declining

Updated @ 00:45 UTC (July 67)

Noteworthy Flare Events (Past 72h)

M5.3 AR 4479 7/5/26 @ 17:57 UTC

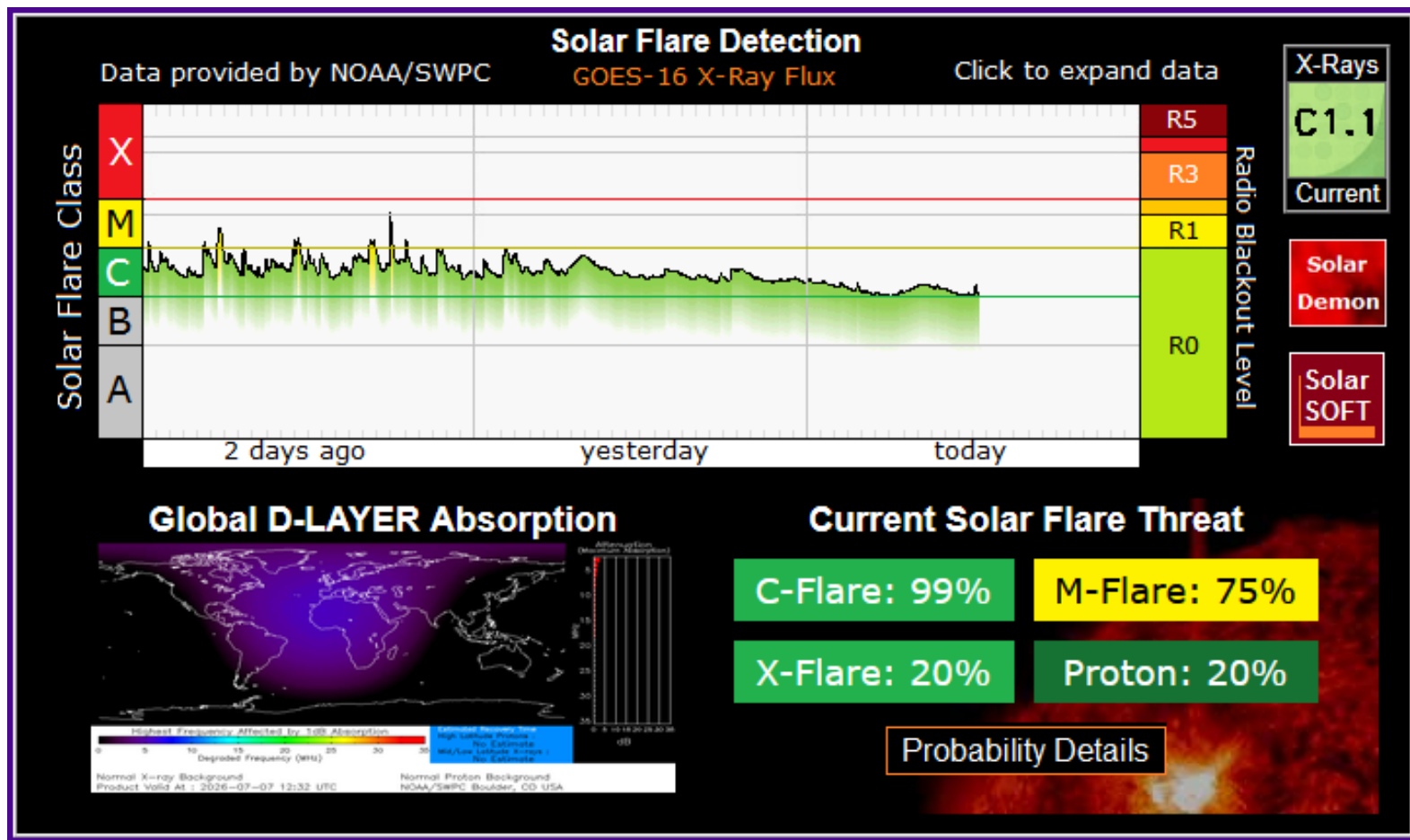
X1.3 E Limb 7/4/26 @ 20:41 UTC

Type II RE (2714 km/s) DIM
10cm Radio Burst (7m, 890 sfu)

M3.2 AR 4479 7/4/26 @ 13:51 UTC

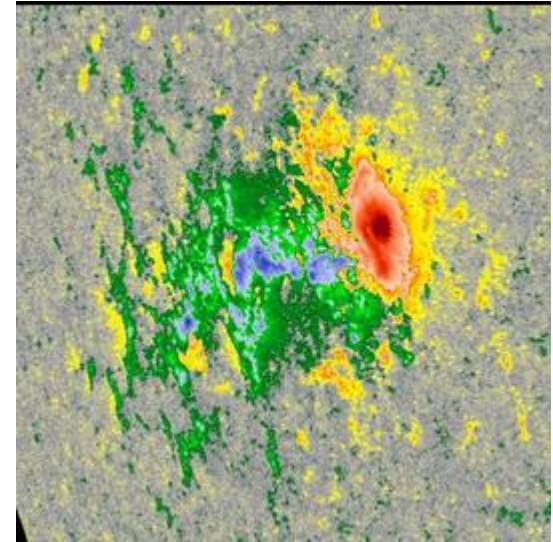
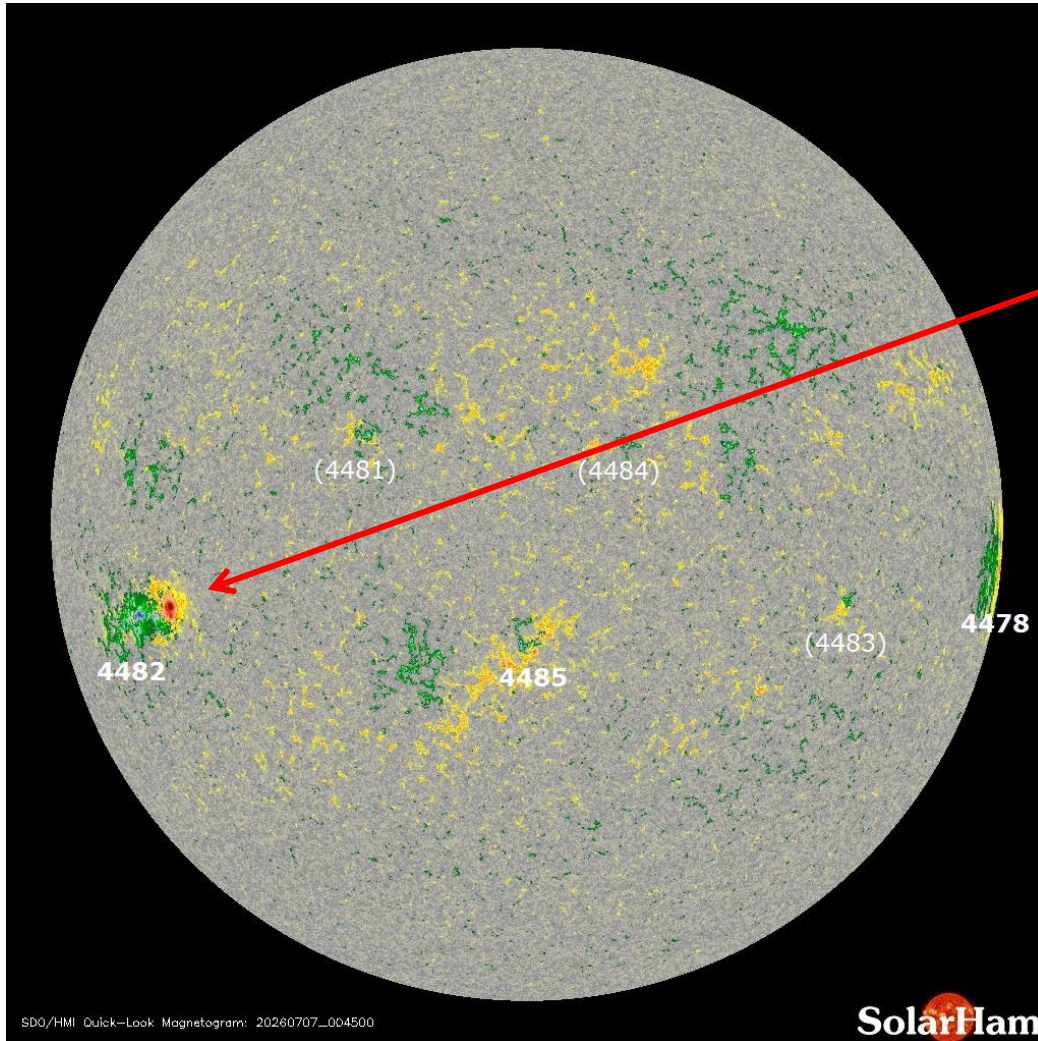
M3.2 AR 4479 7/4/26 @ 11:12 UTC

SolarHam



Sun Spots

Magnetogram Image (Updated July 7, 2026)

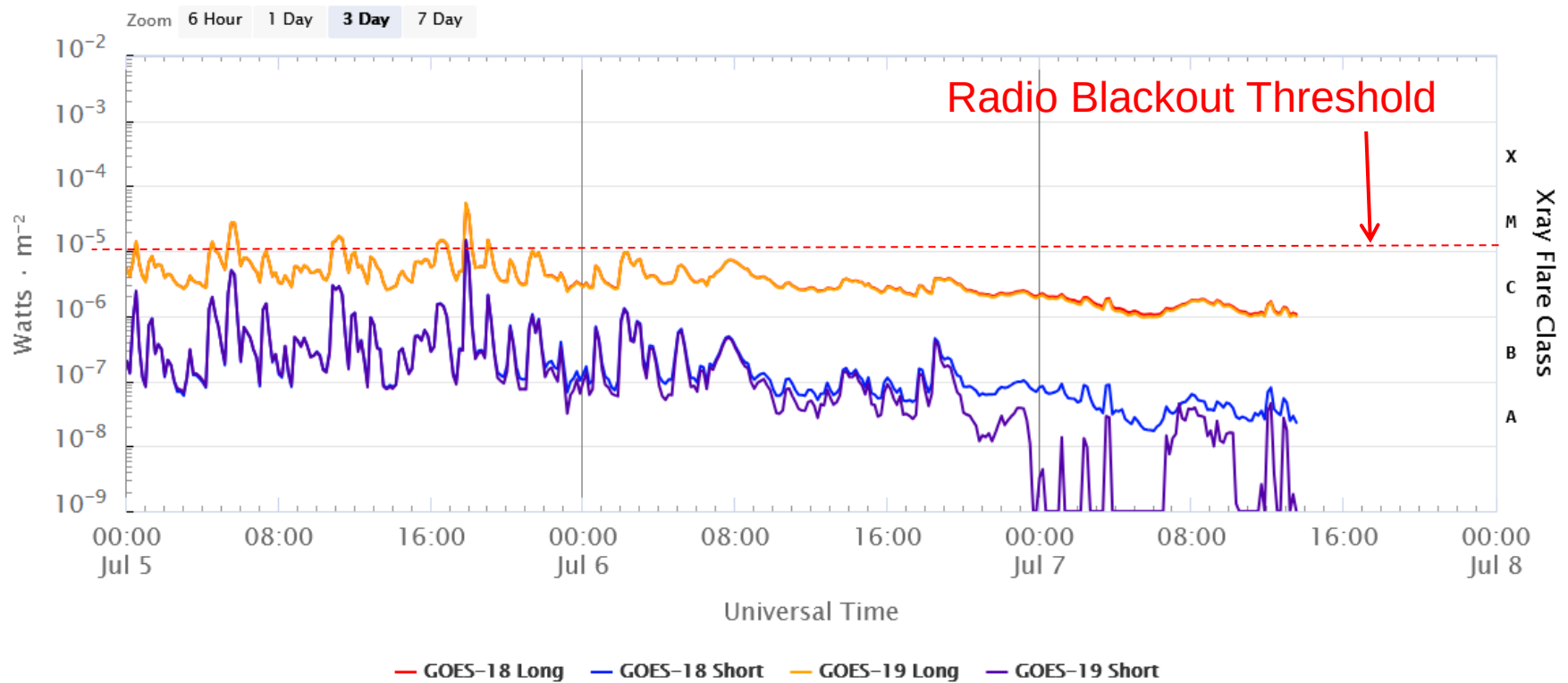


Beta Gamma

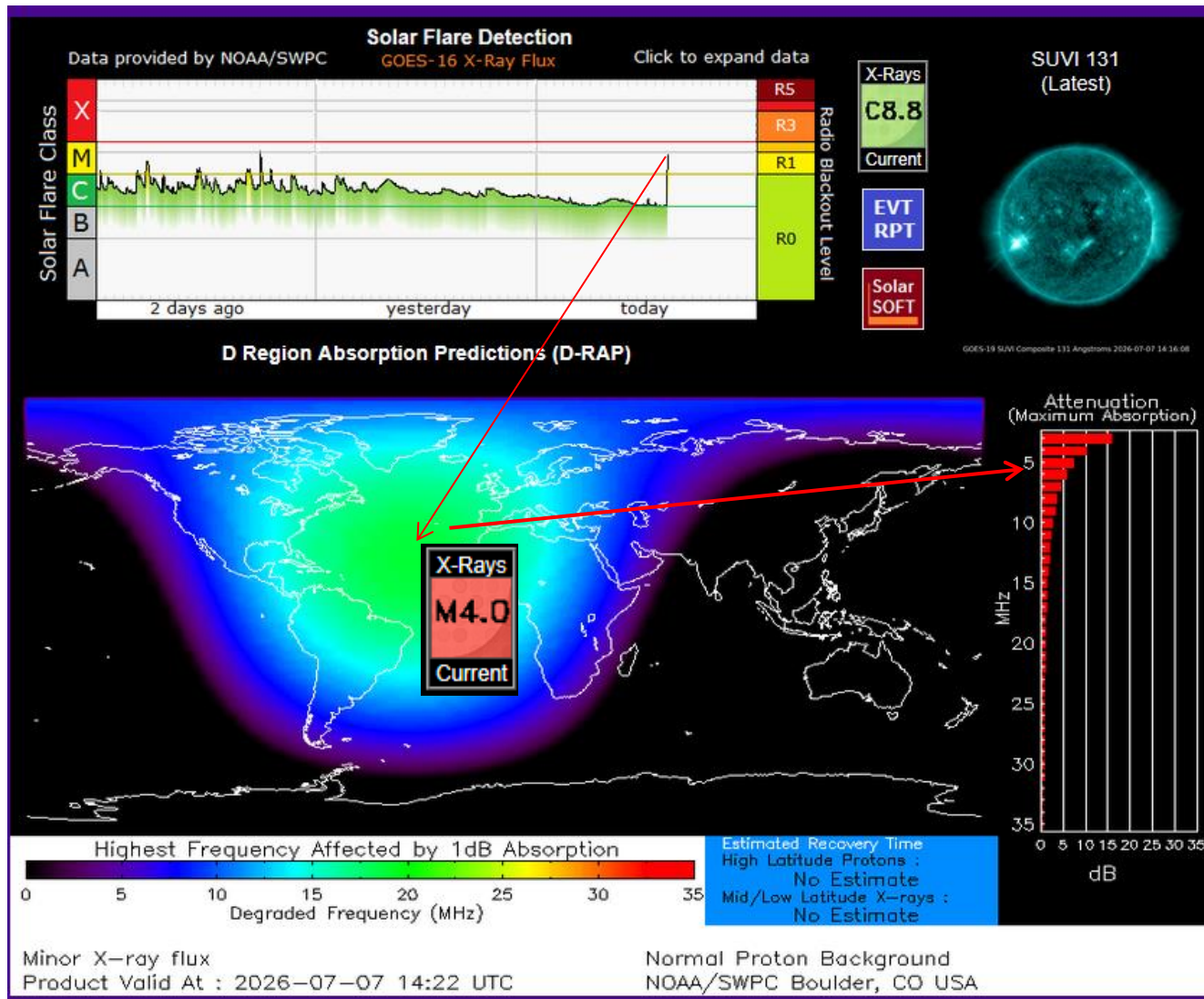
Noteworthy Events

X1.3 (7/4/2026 @ 20:41 UTC)

Solar X-Ray Flux: 5 – 7 JUL 2026

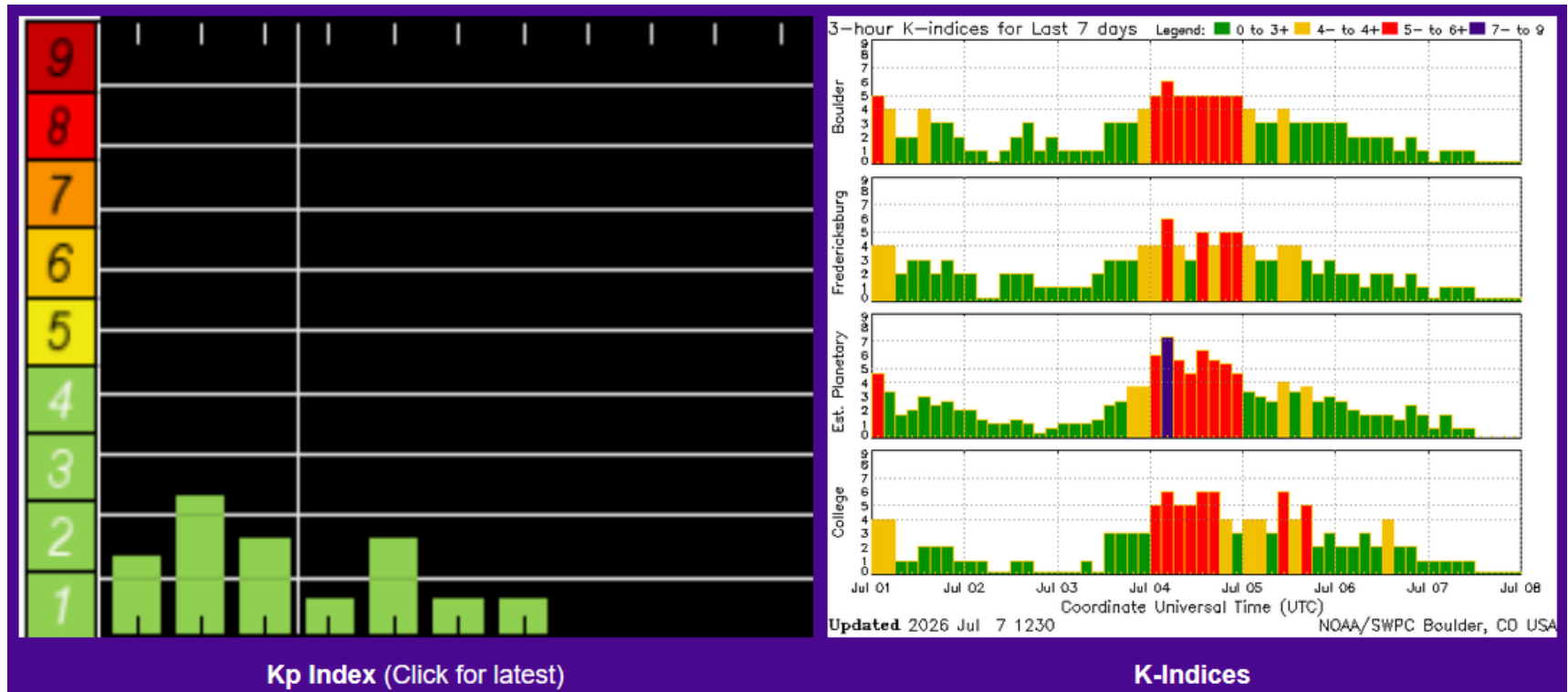


NOAA – D-Region Absorption Predictions



Earth's Geomagnetic Activity

7 JUL 2026



Geomagnetic Conditions: 7 JUL 2026

Solar wind:

$B_z = -0.36$ nT

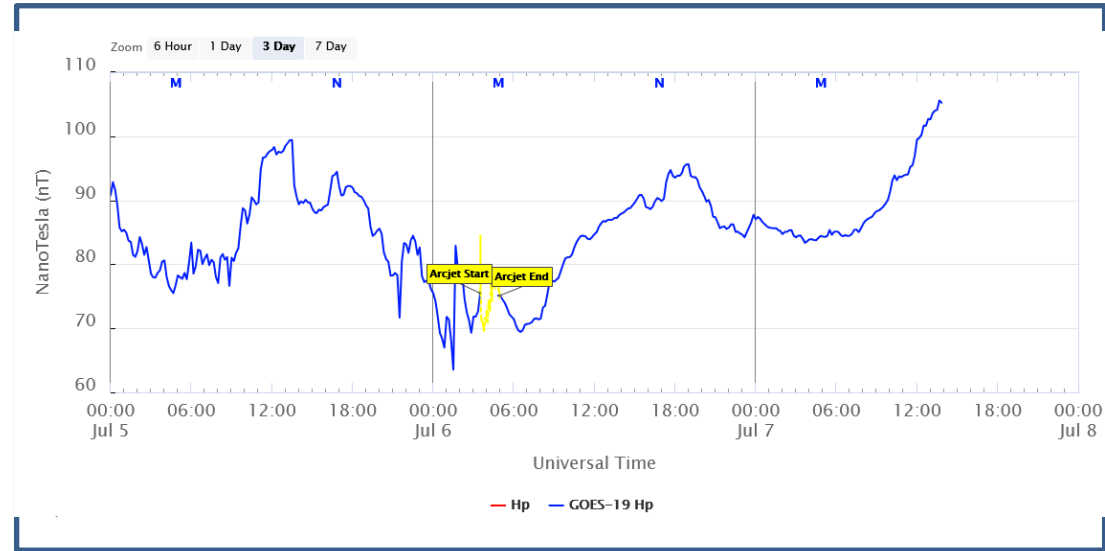
speed = 427 km/sec

density = 0.36 protons/cm³

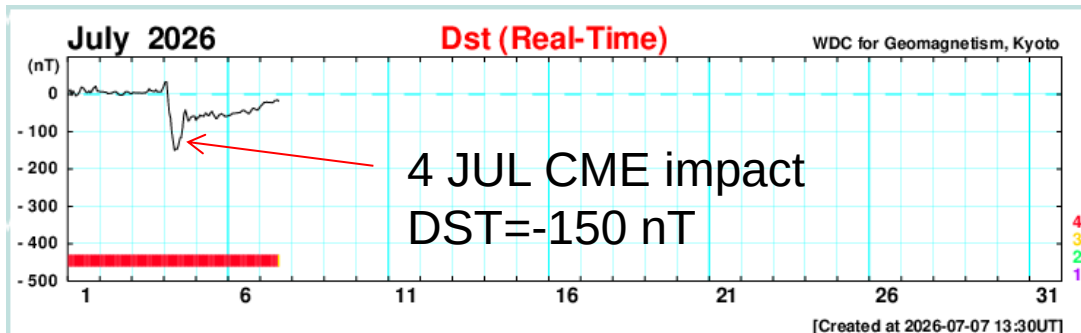
(From – NOAA DSCOVR
In L1, Lagrange Point)

Dst = -15 nT (Ring Field)

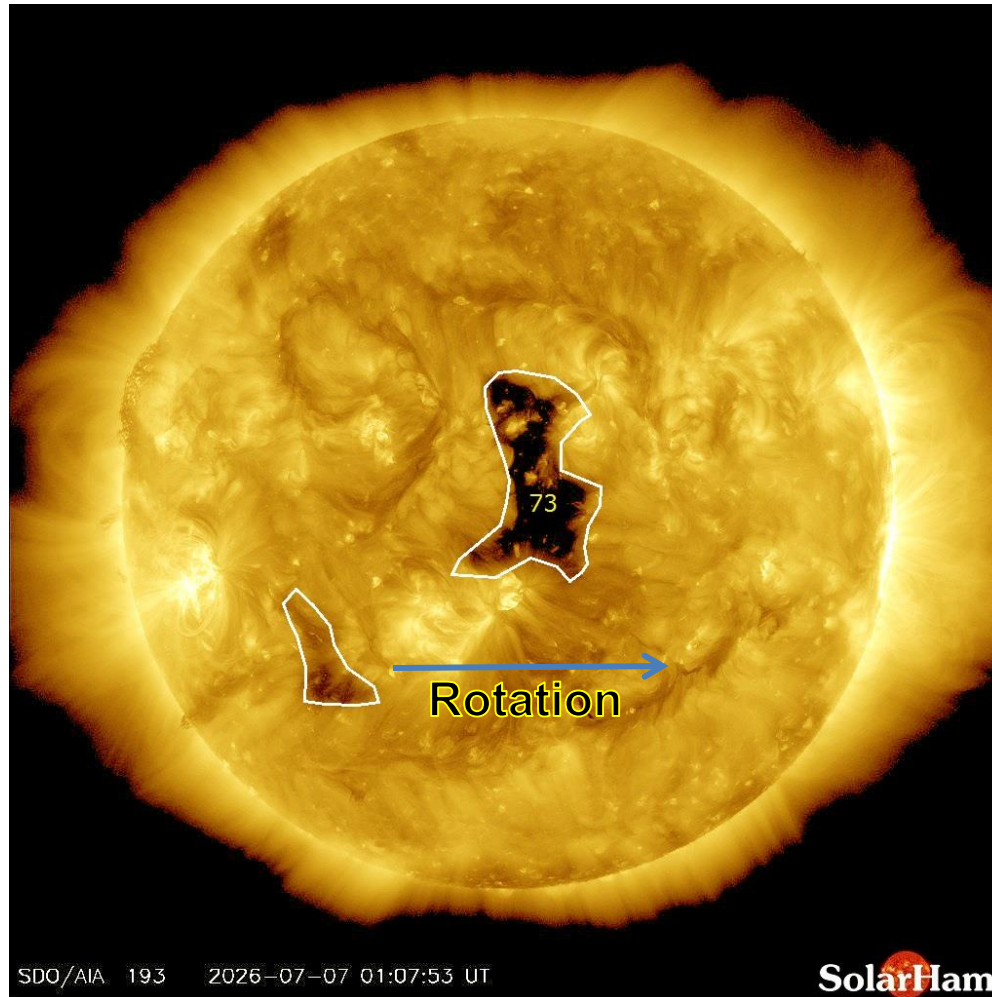
(From – Data Analysis Center
For Geomagnetism and Space
Magnetism – Kyoto University)



From – GOES 19
In geostationary orbit



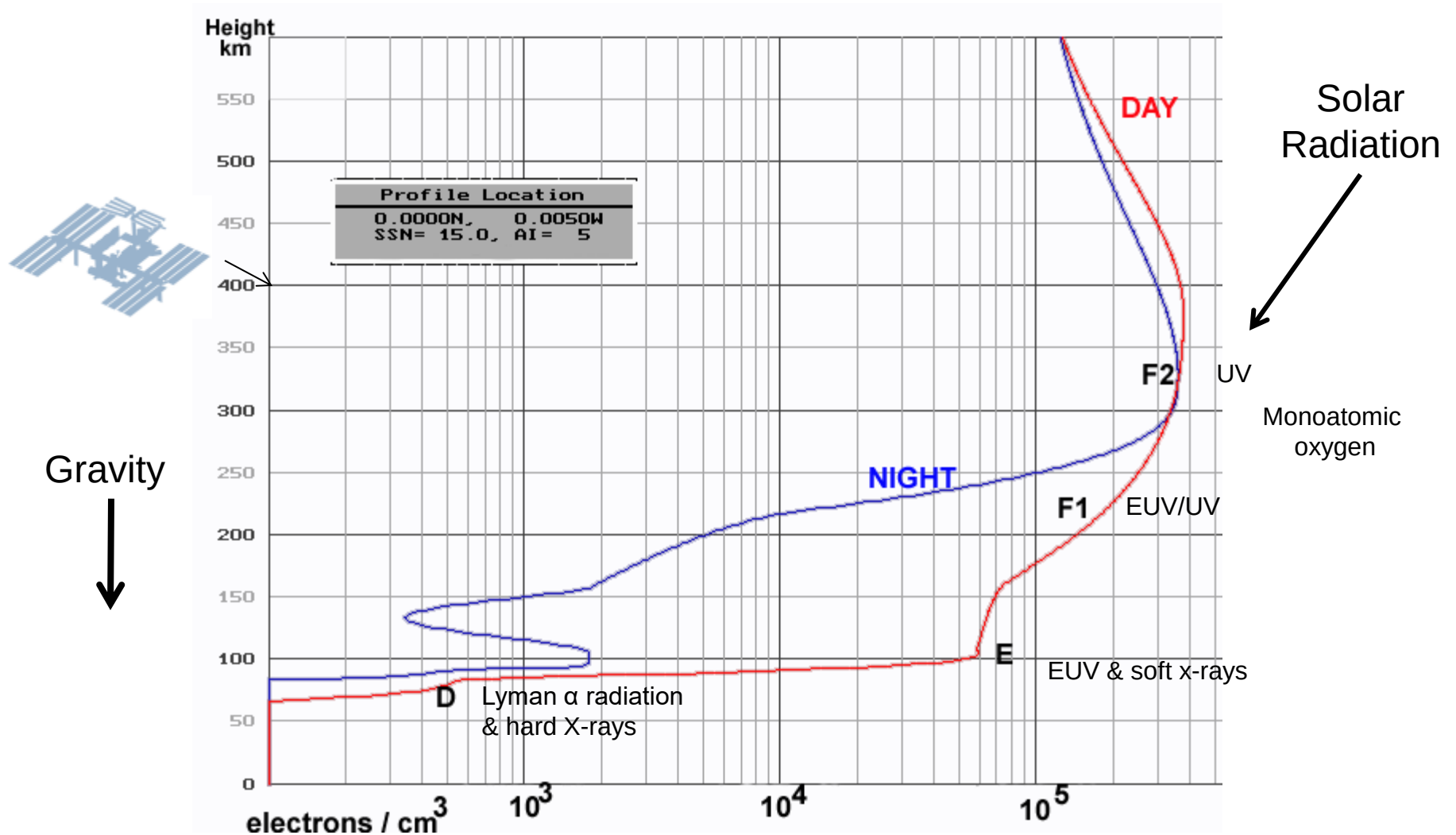
Coronal Holes – 7 JUL 2026



Analysis

Coronal hole #73 is now beginning to directly face Earth. A solar wind stream flowing from this zone should reach Earth by July 9th.

Ionosphere Creation

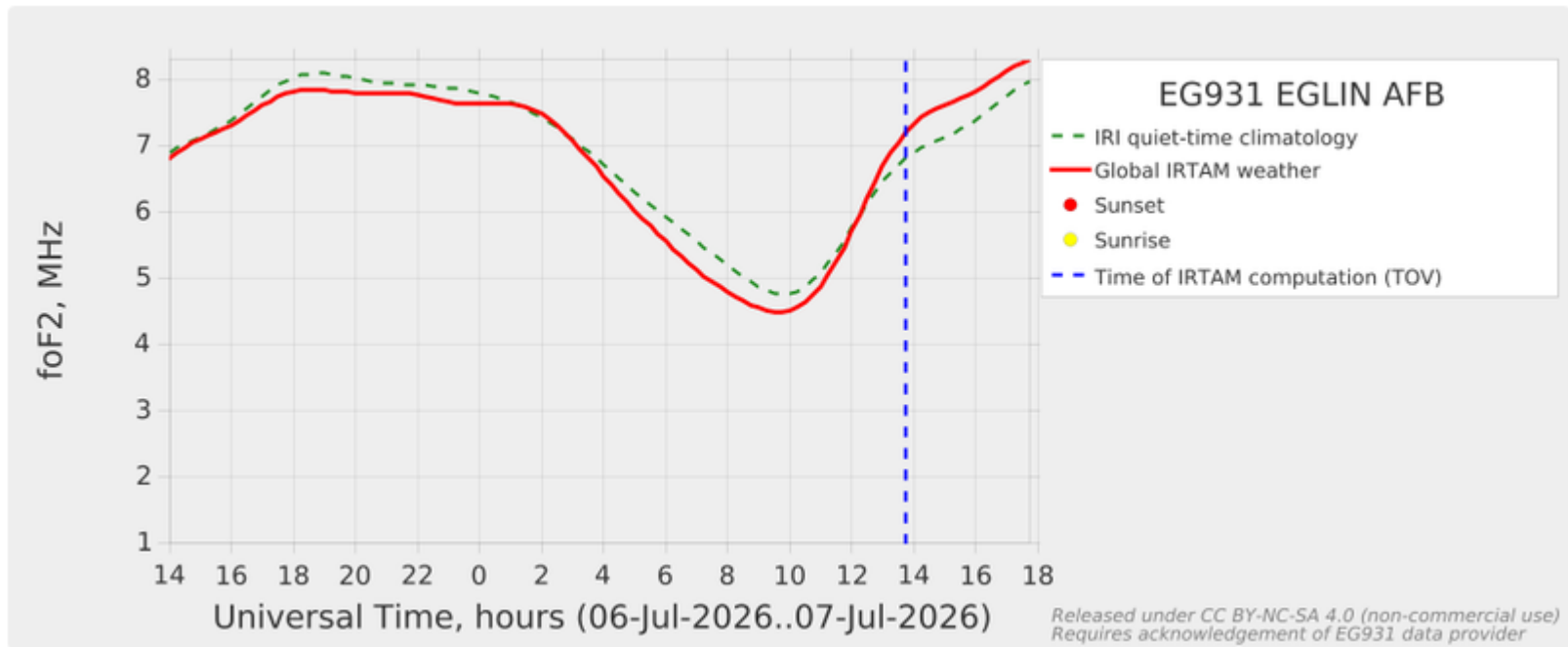


Austin Ionosonde Status

- GIRO is presently doing a software rebuild but GAMBIT is working for Eglin AFB Ionosonde. Austin Ionosonde was heard sounding at my QTH, but Austin GAMBIT plots are modeled only, but appear to be accurate, based on Eglin plot and MARS net performance (Critical Frequency).

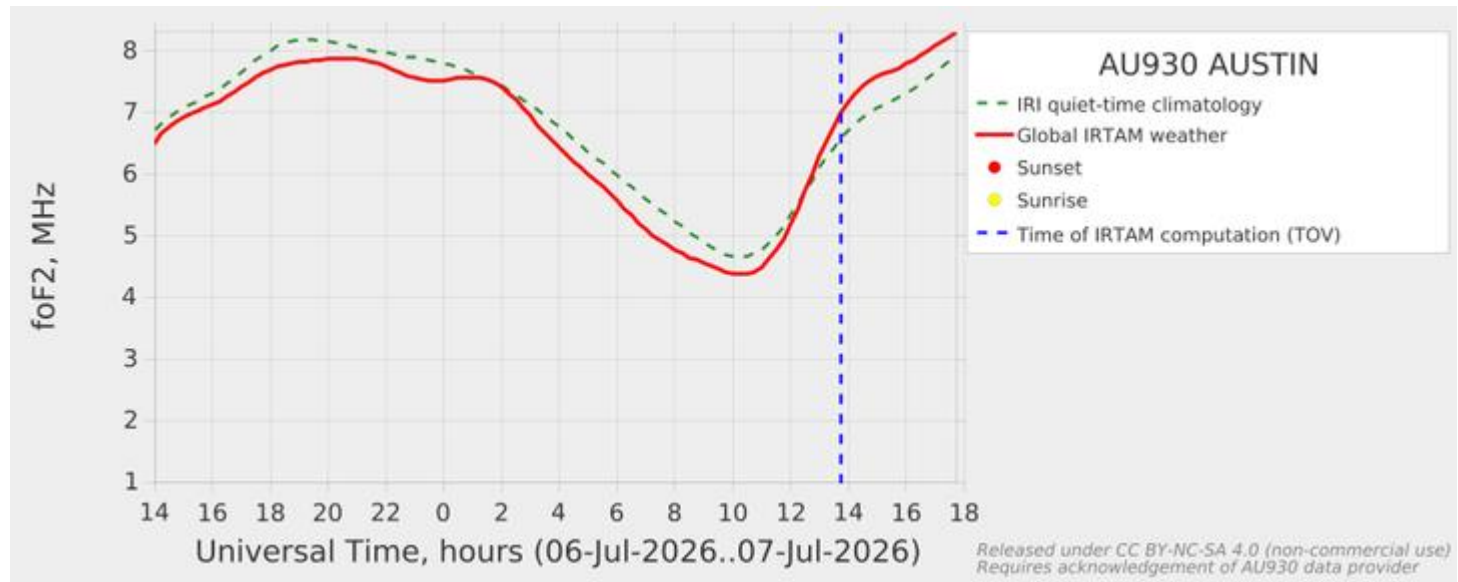
GAMBIT foF2 Trending Chart for Eglin Ionosonde

<https://www.region6armymars.org/resources/solarweather.php>



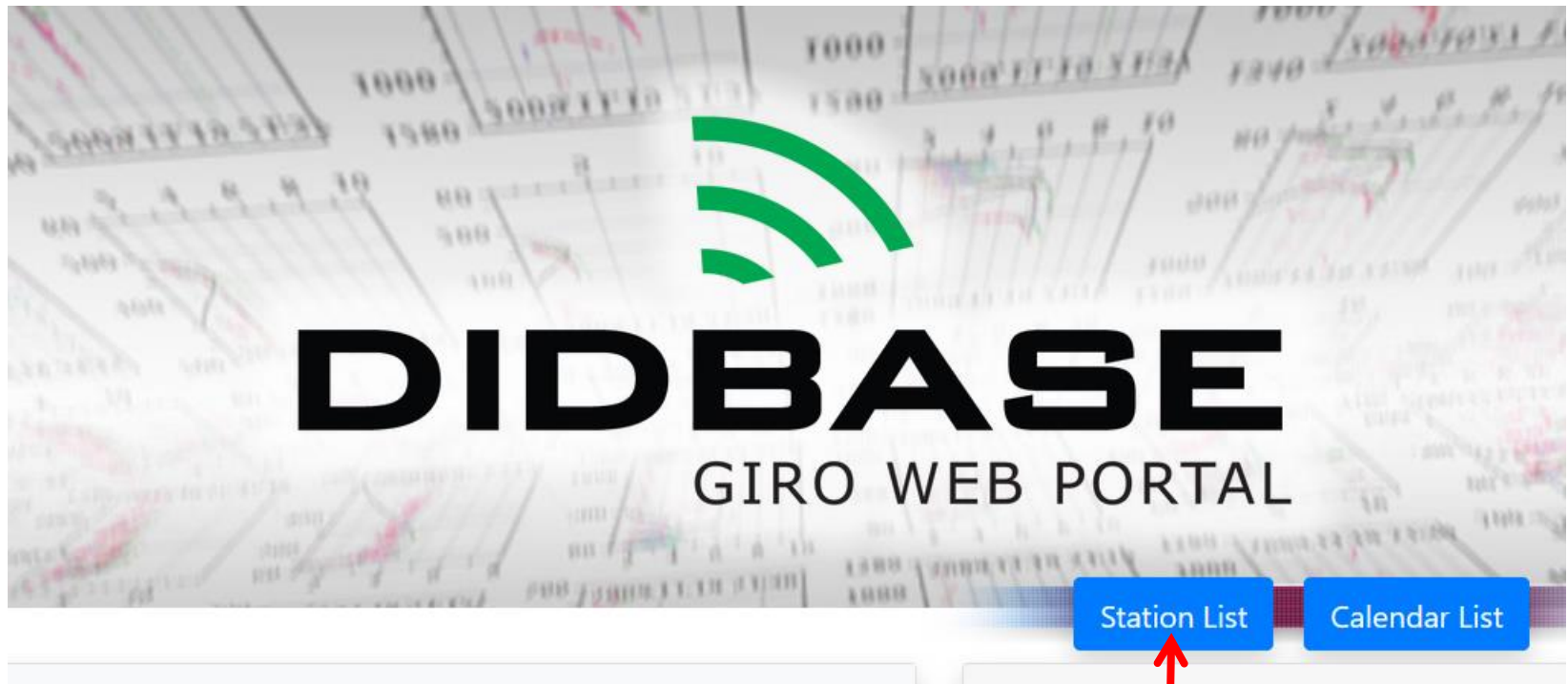
GAMBIT foF2 Trending Chart for Austin Ionosonde – Model only

<https://www.region6armymars.org/resources/solarweather.php>



Use of GIRO DIDBASE

<https://giro.uml.edu/didbase/>

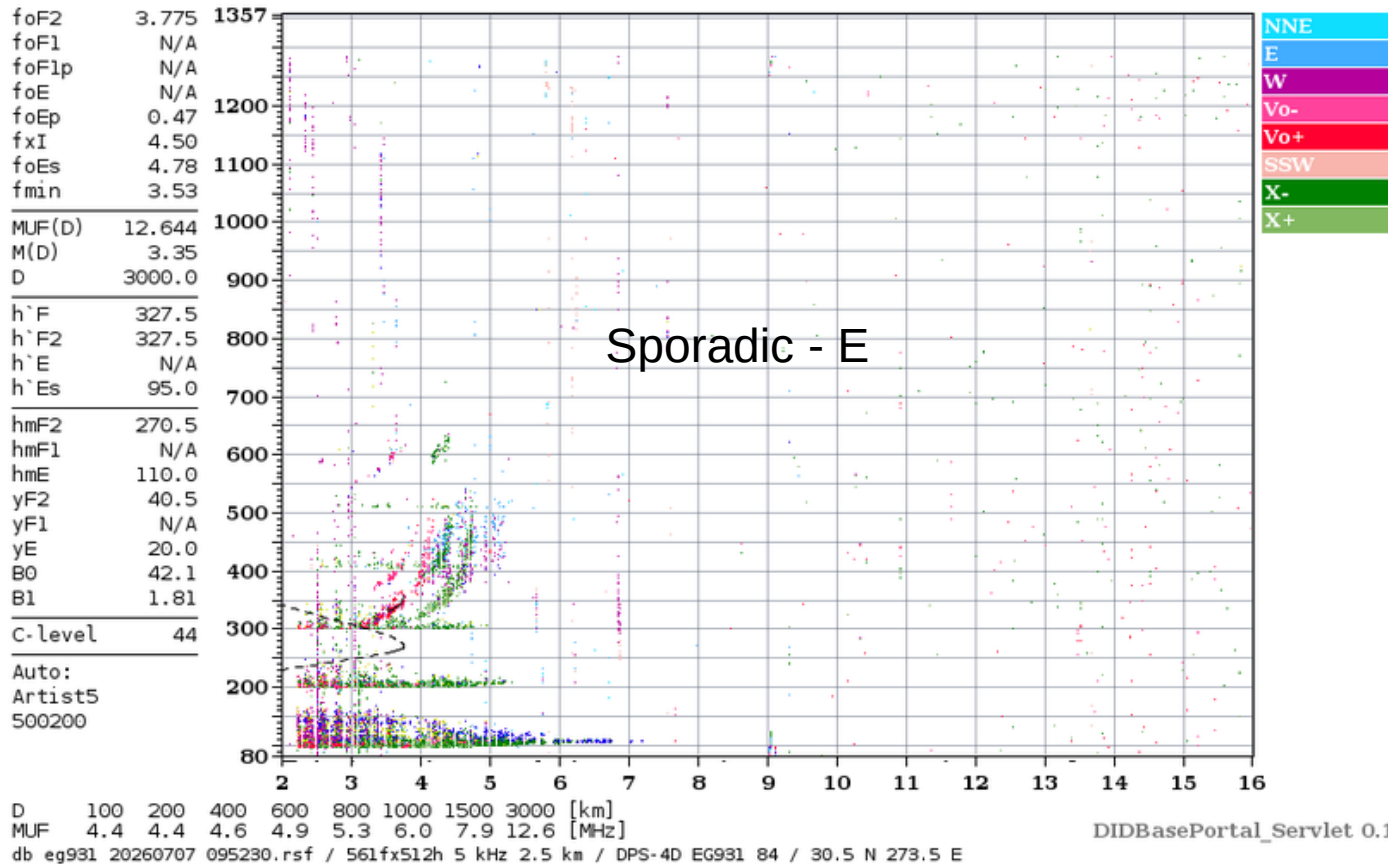


Eglin AFB

Eglin Ionosonde – 0952 DST

Lowell GIRO Data Center

Station YYYY DAY DDD HHMMSS P1 FFS S AXN PPS IGA PS
Eglin AFB 2026 Jul07 188 095230 RSF 1 712 100 03+ C0



Time Shift for Eglin Ionosonde

- Eglin AFB Ionosonde is east of Austin at about the same Latitude so sun angle is about the same (same solar insolation).
- What happens to the Ionosphere over Eglin will occur over Austin about 45 minutes later.
- This is simply a Longitude difference converted to time.
- So look at Eglin's foF2 45 minutes earlier to see what is going to happen over Austin.
- The most rapid change in foF2 occurs as the sun is rising and setting (morning and evening).

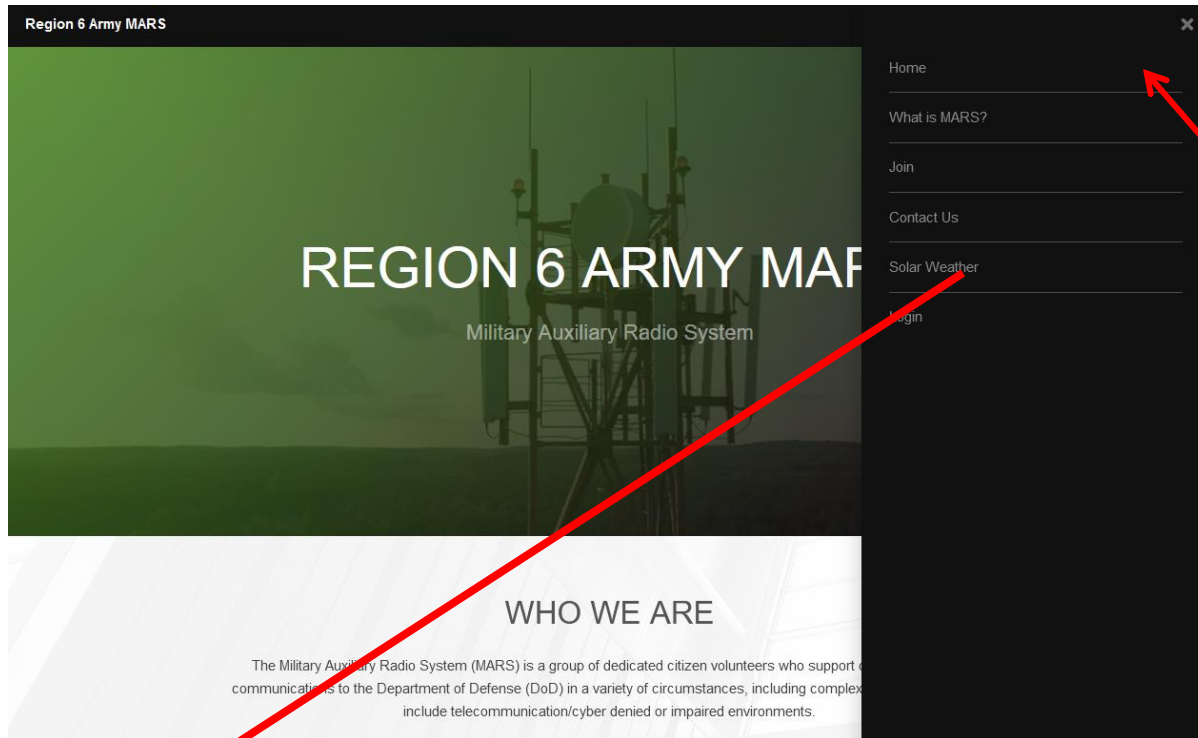
FD 2026 Solar Weather

#			Sunspot		Stanford		GOES15								
#		Radio	SESC	Area		Solar	X-Ray	Flares							
#		Flux	Sunspot	10E-6	New	Mean	Bkgd	X-Ray			Optical				
#	Date	10.7cm	Number	Hemis.	Regions	Field	Flux	C	M	X	S	1	2	3	
#															
2026	06 07	134	124	630	0	-999	*	19	0	0	9	0	0	0	
2026	06 08	131	119	570	1	-999	*	8	0	0	3	0	0	0	
2026	06 09	130	120	760	1	-999	*	18	0	0	6	1	0	0	
2026	06 10	124	114	615	1	-999	*	4	0	0	1	0	0	0	
2026	06 11	127	81	485	0	-999	*	9	0	0	1	1	0	0	
2026	0														
2026	0	Date & Time (UTC) 				Kp Index		Geomagnetic Condition							
2026	0														
2026	0	June 27 (00:00 – 09:00)				2.00 – 2.67		Quiet to Unsettled							
2026	0														
2026	0	June 27 (09:00 – 21:00)				1.33 – 2.00		Quiet							
2026	0														
2026	06 20	113	73	530	1	-999	*	5	2	0	4	0	0	0	
2026	06 21	128	53	280	0	-999	*	4	2	0	3	0	1	0	
2026	06 22	122	81	315	2	-999	*	1	0	0	1	0	0	0	
2026	06 23	130	106	275	2	-999	*	2	0	0	3	0	0	0	
2026	06 24	137	126	640	2	-999	*	12	0	0	8	1	0	0	
2026	06 25	156	136	1175	0	-999	*	10	0	0	8	1	0	0	
2026	06 26	163	99	1210	0	-999	*	5	0	0	10	0	0	0	
2026	06 27	188	119	1480	1	-999	*	9	0	0	25	0	0	0	
2026	06 28	186	131	1500	0	-999	*	7	0	0	8	2	0	0	

FD 2026 Solar Weather

Date & Time (UTC) 	Kp Index	Geomagnetic Condition
June 27 (00:00 – 09:00)	2.00 – 2.67	Quiet to Unsettled
June 27 (09:00 – 21:00)	1.33 – 2.00	Quiet
June 27 (21:00 – 00:00)	3.00	Active
June 28 (00:00 – 09:00)	3.00 – 4.00	Active (Highest peak)
June 28 (09:00 – 18:00)	1.67 – 3.00	Quiet to Unsettled
June 28 (18:00 – 00:00)	2.67	Unsettled

Solar Weather Data



Menu

Solar Weather

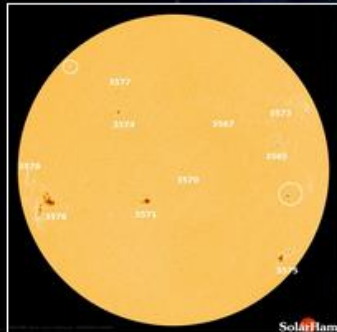
All Ionosondes
GAMBIT URL

- [GAMBIT](#) - Global Assimilative Model of Bottomside Ionosphere Timeline
 - [Boulder](#)
 - [Eglin](#)
- [NOAA Solar Weather](#) - Solar Weather plots of Kp and X-Ray and other solar emissions.
- [Solen Solar Weather](#) - Good general solar forecast from an individual.
- [Solar Ham](#) - SolarHam provides real time solar news, as well as consolidated data from various sources.

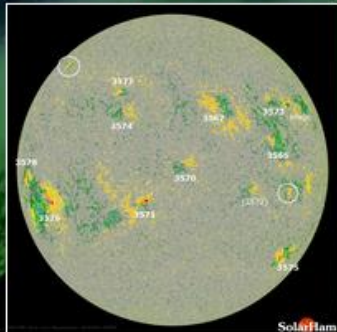
Space Weather for February 6, 2024

[Help Center + FAQ](#)

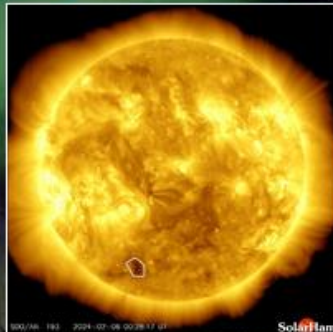
UTC Time 13:45:49 Tuesday



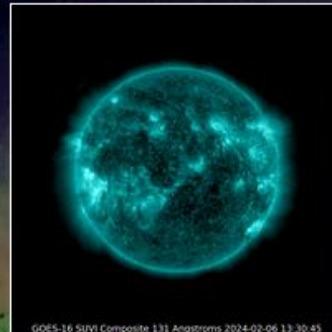
HMI Intensity
[Latest](#) | [Movie](#) | [HARP](#)



HMI Magnetogram
[Latest](#) | [Movie](#)



Coronal Holes
[Analysis](#) | [Movie](#)



SUVI 131 (Latest)
[Movie](#)



SUVI 304 (Latest)
[Movies](#)

Latest Imagery: [SDO](#) | [AIA](#) | [GOES](#) | [GONG](#) | [STEREO](#) | [LASCO](#)

Video: [SDO](#) | [SOHO](#) | [STEREO](#) | [Heliviewer](#) | [YouTube](#)

[Solar Report](#)

[Space Weather Alerts](#) ➔

[Real Time Solar Wind](#)

[Protons and Electrons](#)

[Satellite Environment](#) ➔

Real Time Solar Wind (BETA) | [Expand Data](#)

Speed: 437 km/s

Density: 2.96 p/cm³

Bz: 4.25 nT ↑

Bt: 6.39 nT

Updated every minute.



<https://www.spaceweather.com/>

Current Conditions

Solar wind

speed: **314.8** km/sec

density: **9.9** protons/cm³

more data: [ACE](#), [DSCOVR](#)

Updated: Today at 1225 UT

X-ray Solar Flares

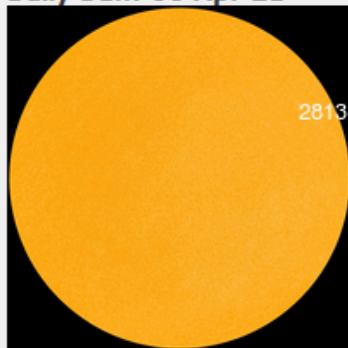
6-hr max: **A1** 1027 UT Apr06

24-hr: **A1** 1515 UT Apr05

[explanation](#) | [more data](#)

Updated: Today at: 1230 UT

Daily Sun: 06 Apr 21



Sunspot AR2813 is decaying, and poses no threat for strong flares.
Credit: SDO/HMI

FLYING TO THE VOLCANO: Iceland's Geldingadalur volcano has turned into an popular tourist attraction—especially since auroras were sighted [above the glowing lava](#). Early this morning, Tuesday, April 6th, Brian Emfinger saw auroras before he even reached the Reykjanes peninsula:



QUESTIONS?

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